



Emergency Vehicle Priority Routing

TEAM - 15

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The Problem: A Life-or-Death Metric

Urban congestion is a critical bottleneck, threatening the efficacy of emergency services, especially within the critical "Golden Hour" for medical emergencies.

The "Golden Hour" Constraint

For ambulances, every minute lost to traffic significantly reduces the chance of patient survival or recovery.

Archaic Preemption Systems

Current systems (e.g., optical preemptors) are not adaptive to real-time flow and lack network-wide coordination capabilities.

Failure in Conflict

These traditional systems are incapable of handling multiple, simultaneous high-priority requests for the same intersection.



The Core Challenges: Defining the System Requirements

Implementing an effective EV priority system requires overcoming three primary technical hurdles:



Real-Time Detection & Fusion

Minimizing latency in reliably identifying the EV, verifying its priority level (e.g., Code Black), and communicating status.



Dynamic Pathfinding

Calculating the optimal, least-congested route on the fly, as network conditions change in real-time.



Priority Task Scheduling

The system's core challenge: How to effectively de-conflict resources when multiple high-priority EVs compete for the same intersection or route segment.



Our Solution: The Priority Routing System

Guardian Flow leverages a hybrid model, combining Edge Computing for speed and Computational Intelligence for coordination.

Edge Computing: The Instant Reflex

Enables instantaneous, sub-second decisions right at the intersection for immediate response.



Computational Intelligence: The Intelligent Swarm

Facilitates smart, coordinated, and resource-scheduled decisions across the entire traffic network.

Key Operational Outcomes

"Green Corridor" (Local Action)

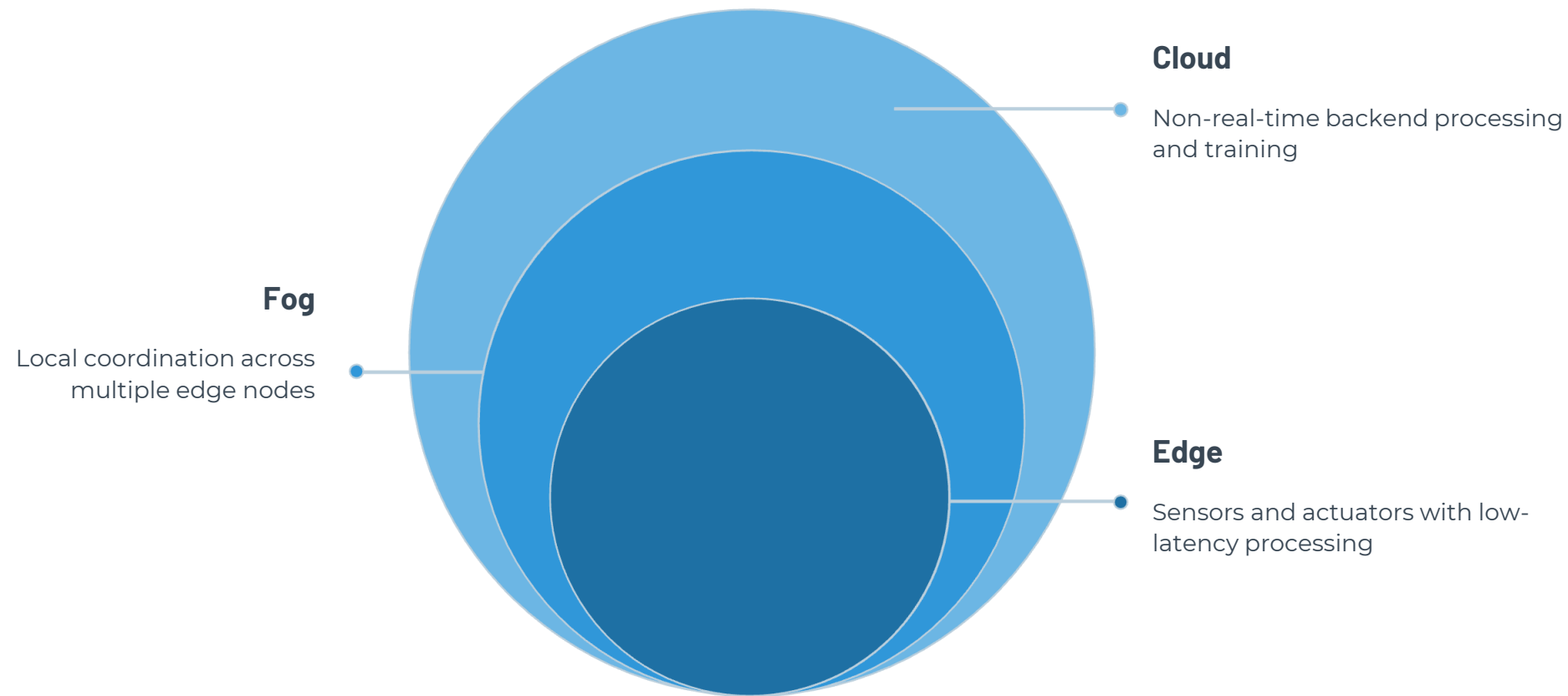
An instantaneous, localized action to clear the EV's current lane at a single intersection by immediately controlling the traffic light.

"Green Wave" (Network Action)

A coordinated, network-level plan executing a sequence of synchronized green lights along the calculated optimal path.

Proposed 3-Layer Architecture for Distributed Computation

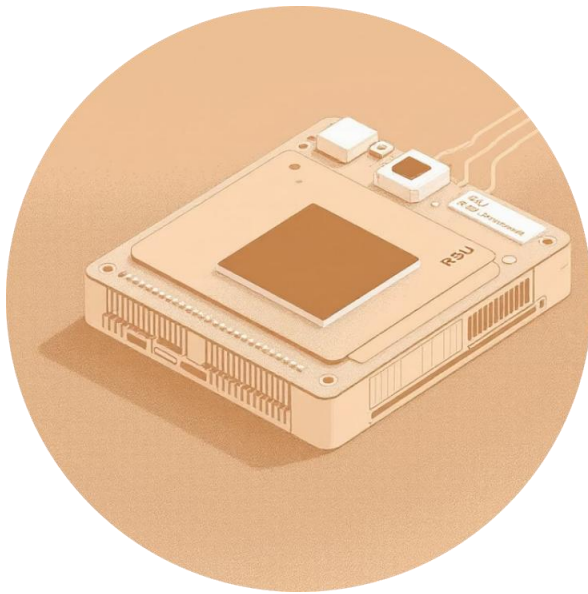
The Guardian Flow system distributes processing and decision-making across three distinct layers to ensure optimal balance between latency, coordination, and training capability.



This hierarchical structure ensures that critical real-time decisions happen locally (Edge), coordination occurs nearby (Fog), and heavy computation is offloaded (Cloud).

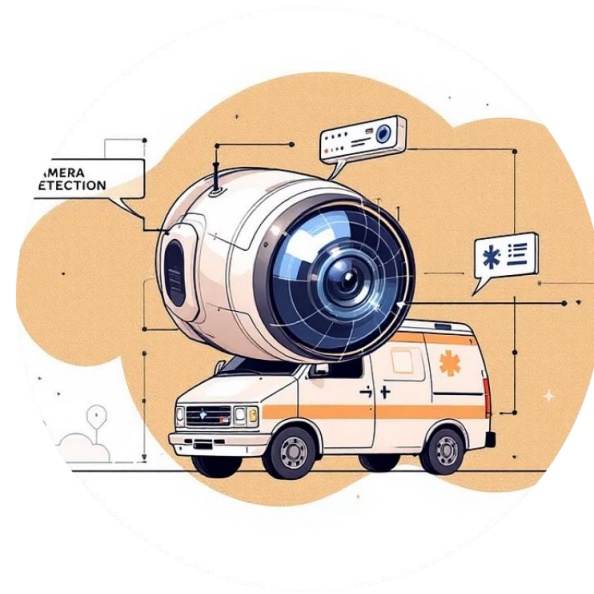
Layer 1: The Edge (The Instant Reflex)

The edge layer is designed for low-latency detection and immediate local action to clear the path for the emergency vehicle.



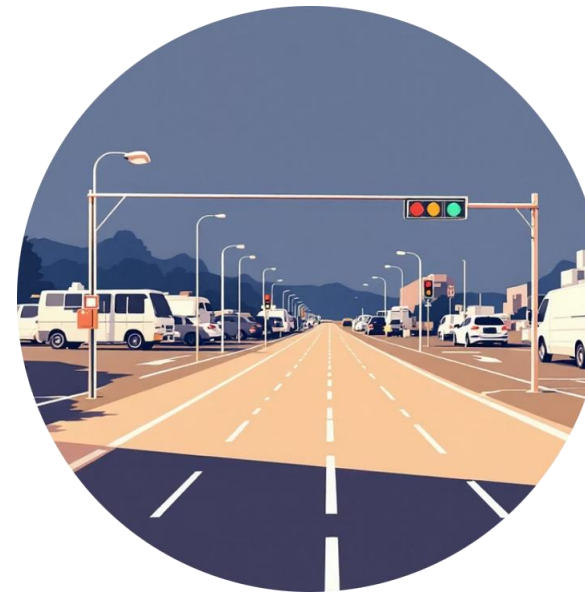
Hardware at the Intersection

Deployed on resource-constrained devices like a Jetson Nano, paired with a Road-Side Unit (RSU).



Detect & Fuse Data

Utilizes YOLOv8 for visual detection, which is fused with the official priority code ("Code Black") transmitted by the EV's On-Board Unit (OBU).



Immediate Actuation

Instantly executes the "Green Corridor" local model by overriding the traffic signal state for sub-second response.



Alert Fog Layer

After actuation, it sends the detection and action data to the Fog Layer for network-wide coordination.



Layer 2: The Fog (The Intelligent Swarm)

The fog layer forms a decentralized network responsible for network-wide coordination and resolving resource conflicts.

Decentralized Network

The Fog is a collective of interconnected intersection agents, operating the core logic using Computational Intelligence (CI).

Optimal Pathfinding

Continuously uses real-time congestion and priority data from all edge agents to calculate and update the lowest-cost, optimal network path.

Priority Task Scheduling

Manages a dynamic priority queue to solve the core challenge: de-conflicting simultaneous requests from multiple EVs for the same route segment.

Green Wave Coordination

Commands the full "Green Wave" sequence, dictating the synchronized timing of subsequent traffic lights along the EV's chosen route.

Layer 3: The Cloud (The AI Factory)

The Cloud serves as the non-real-time backend, dedicated to continuous improvement and long-term analysis.



AI Model Training

Leverages powerful, centralized GPU resources to train, optimize, and improve the accuracy of the YOLO object detection model used at the Edge.



Data Warehousing

Acts as a central repository, storing historical logs and traffic data aggregated from all intersection agents.



Long-Term Analytics

Performs complex analysis on stored logs to identify traffic patterns, refine system parameters, and generate urban planning insights.

Methodology & Virtual Prototype

We validated the Guardian Flow system using a robust simulation environment.

Edge Logic: "Green Corridor"

- Simple, robust rule-based logic.
- Immediately sets all conflicting cross-traffic lanes to RED.
- Sets the EV's entire lane to SOLID GREEN to ensure the path is instantly cleared of any blockages.

Swarm Logic: "Traffic Police"

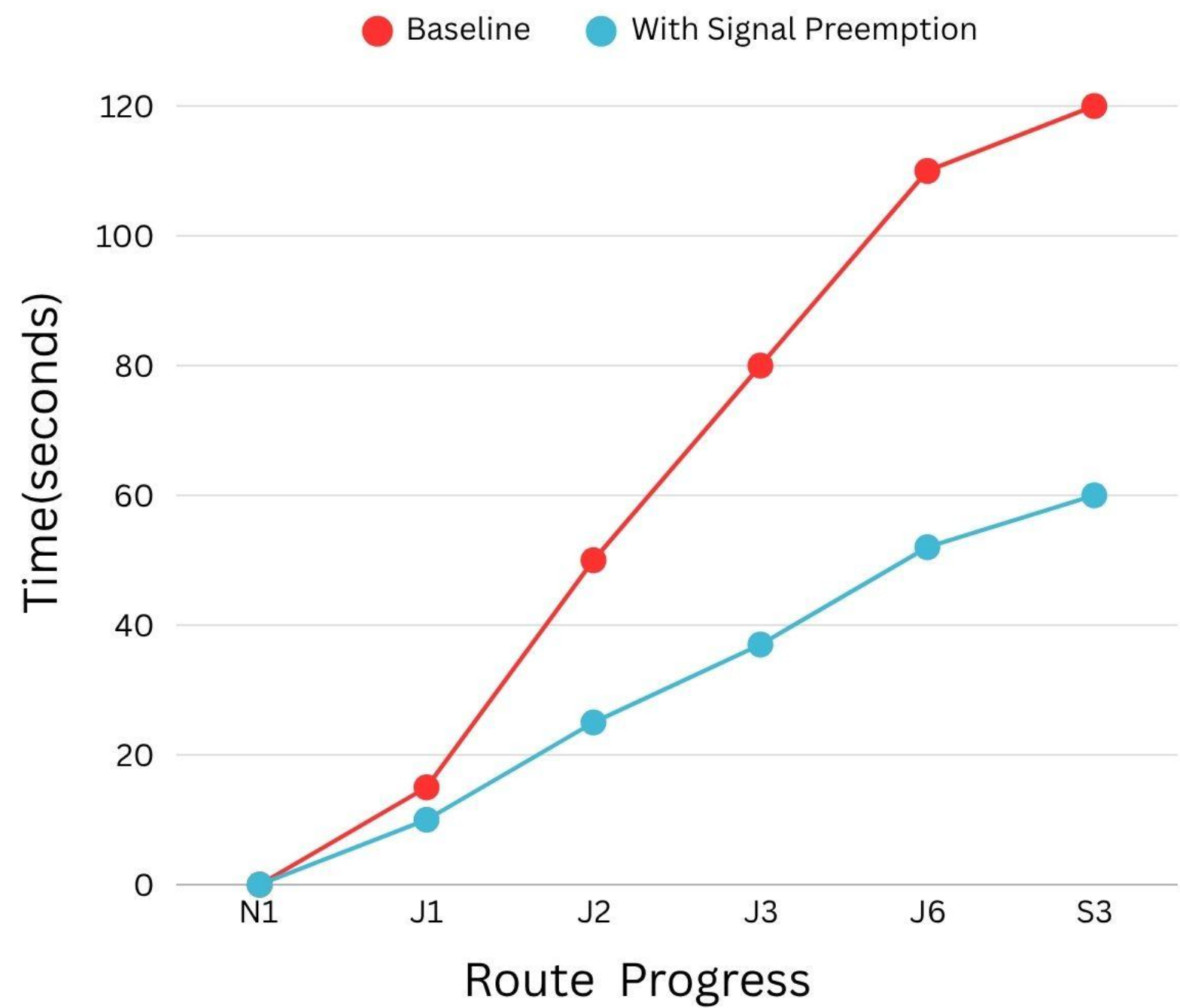
- Adaptive network logic based on real-time congestion levels.
- Includes phases like "Clear Out" for intersections with low cross-traffic volume.

Simulation Environment

- Built using **SUMO** (Simulation of Urban MObility) for realistic traffic modeling.
- Controlled by a Python interface (TraCI) which acts as the intelligent controller, replicating our Fog layer logic.



Results: Ambulance Travel Time Comparison



Results: Optimal Path from swarm agent

🌀 Time step 1 -----

[Central Node] Global average congestion: 41.80
🚦 Best route at t=1: [1, 3, 4, 5] | Cost: 91.00

🌀 Time step 2 -----

[Central Node] Global average congestion: 43.20
🚦 Best route at t=2: [1, 3, 2, 4, 5] | Cost: 196.00

🌀 Time step 3 -----

[Central Node] Global average congestion: 35.00
🚦 Best route at t=3: [1, 3, 4, 5] | Cost: 77.00

🌀 Time step 4 -----

[Central Node] Global average congestion: 32.60
🚦 Best route at t=4: [1, 2, 4, 5] | Cost: 115.00

🌀 Time step 5 -----

[Central Node] Global average congestion: 31.40
🚦 Best route at t=5: [1, 3, 2, 4, 5] | Cost: 147.00

✅ Simulation complete.